Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team details

Ember

Turns what your kids watch into what your family does

Team name
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Team leader
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Problem statement

Children watch hours of video a day, and much of it is good. But parents only see the hours. Nothing tells them what a child watched, whether it was worth the time, or what to do about it.

Brief about the idea

Ember turns what your child watches into what your family does

Parents can see how long a child watched, but not what the videos contained or whether they were worth the time. Ember reads the videos a child likes on YouTube and has Gemini watch each one. The parent gets a plain report, a map of what draws the child, and one weekend activity to do away from the screen.

Know what they watched

A summary, learning value and safety note for every video, so nobody has to rewatch.

See what draws them

Videos grouped by what holds attention, not by topic. Ramen and robots can share a theme.

Do something about it

One printable activity each week, built from their strongest interest.

Opportunities

How is it different from existing ideas?

Screen-time tools restrict by hours, and YouTube's recommender aims for the next video. Tools that read titles cannot tell a survival documentary from a survival game stream. Ember reads what is inside the video, and ends offscreen.

How will it solve the problem?

Gemini watches each liked video and writes what happens, what a child may learn, and anything a parent should know. A dashboard rolls it up into one view, so a parent can see the pattern and act on it.

What is the USP?

Video understanding, not metadata. Themes that carry reasoning and evidence, not tags. Every output ends with something to do offscreen: a printable Quest Card.

Tested on 17 real liked videos: the food was incidental. What held attention was things made with visible precision, which also explains a cleaning robot sitting beside the ramen.

Features offered by the solution

Understand

Gemini watches the video

Reads what happens in each liked video, not just the title or tags.

Report

A report for every video

Plain summary, what a child may learn, age fit and how educational it is.

Alert

A worth-a-look list

Pushy ads, risky challenges and misleading claims come first, with a question to ask.

Map

What draws your child

Videos grouped by the pull behind them, with evidence, trend and confidence.

Act

One weekend Quest Card

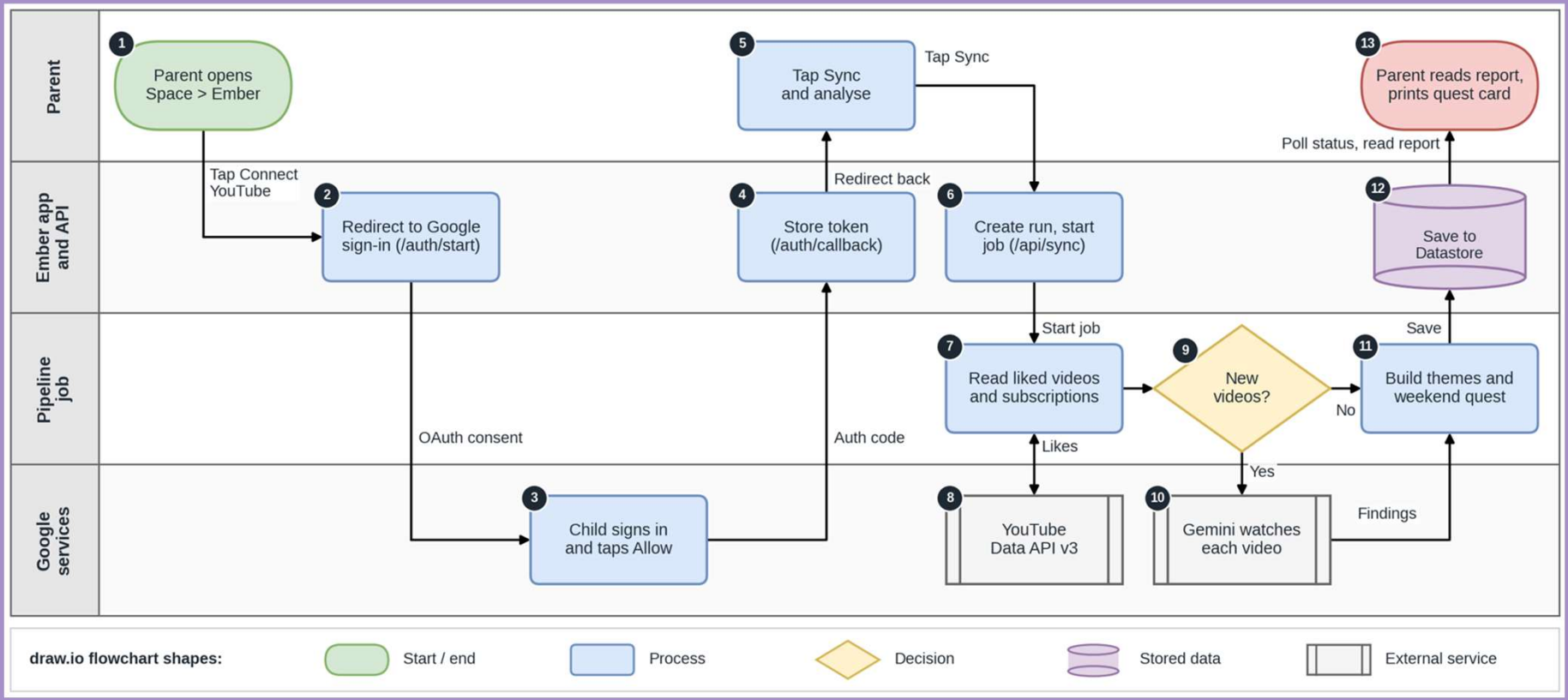
A doable offline activity: who leads, who helps, printable.

Connect

One tap, or paste links

Read-only access to liked videos, or save links by hand. No file exports.

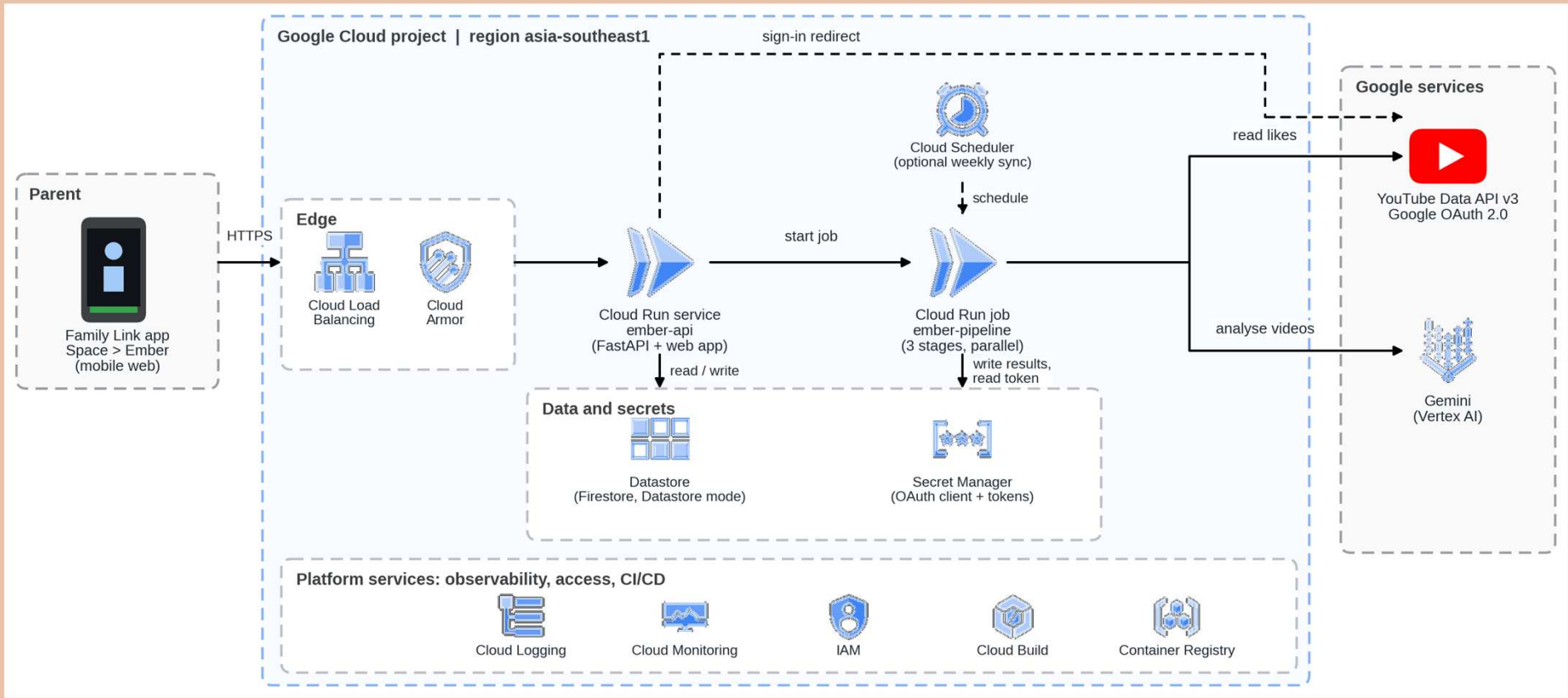
Process Flow Diagram



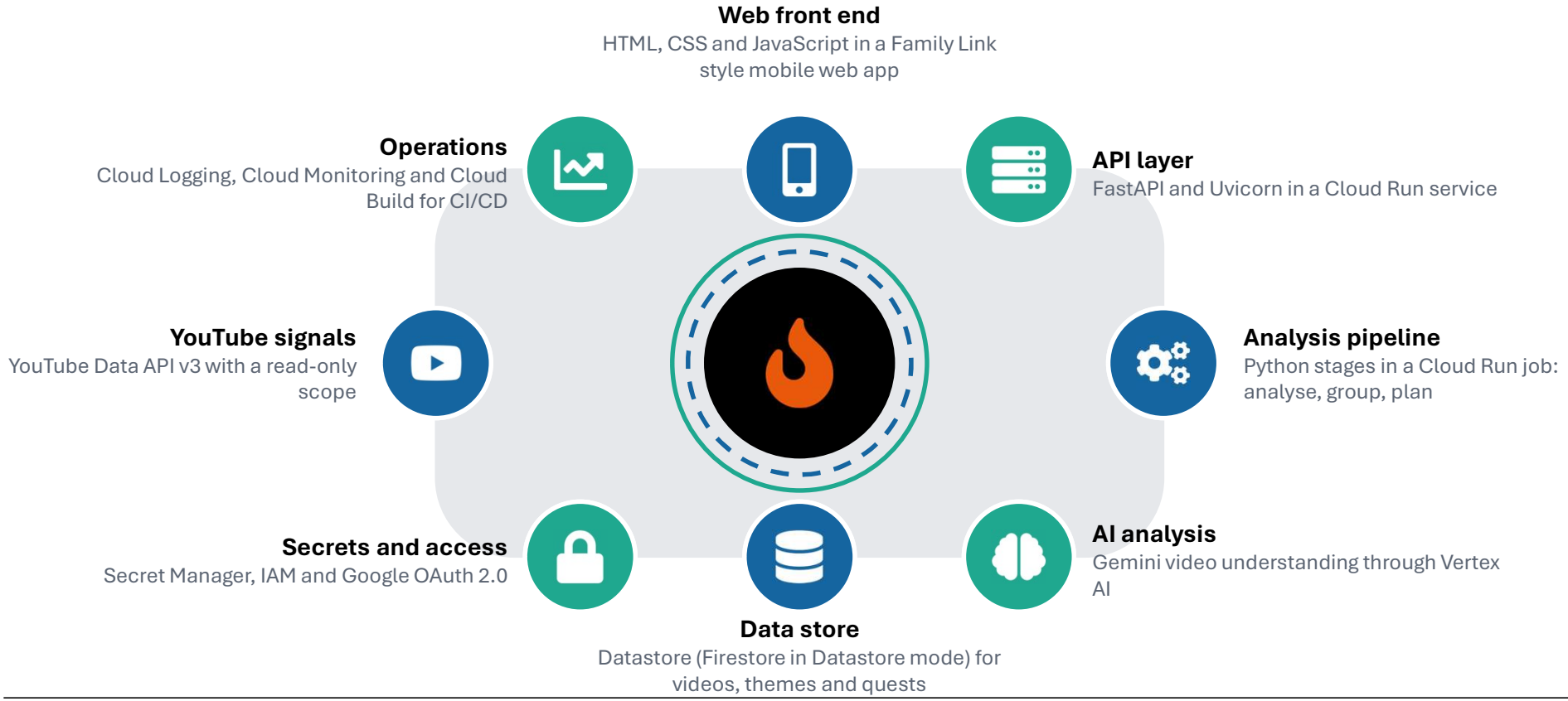
Turns what your kids watch into what your family does

Ember

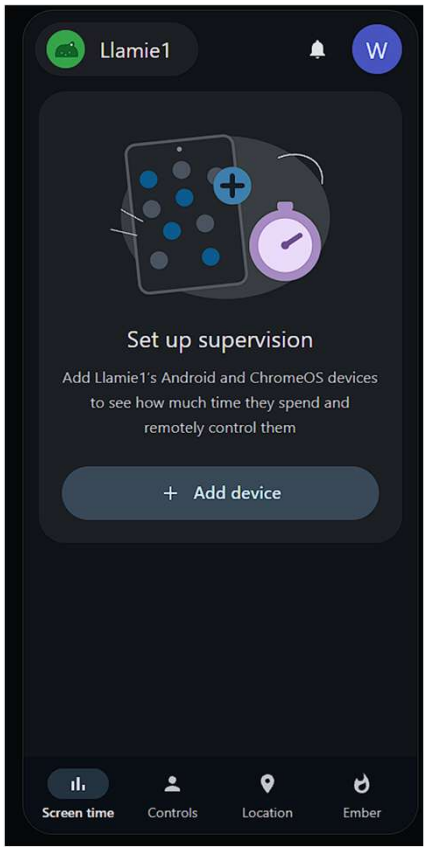
Architecture Diagram

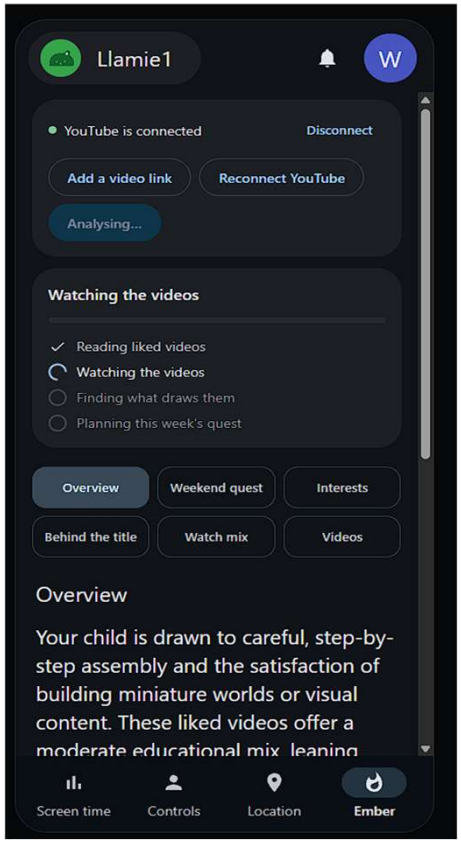


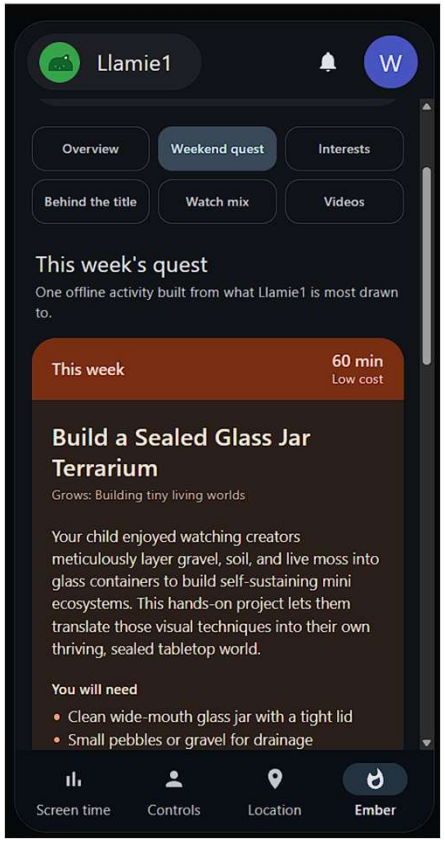
Technology used in the solution

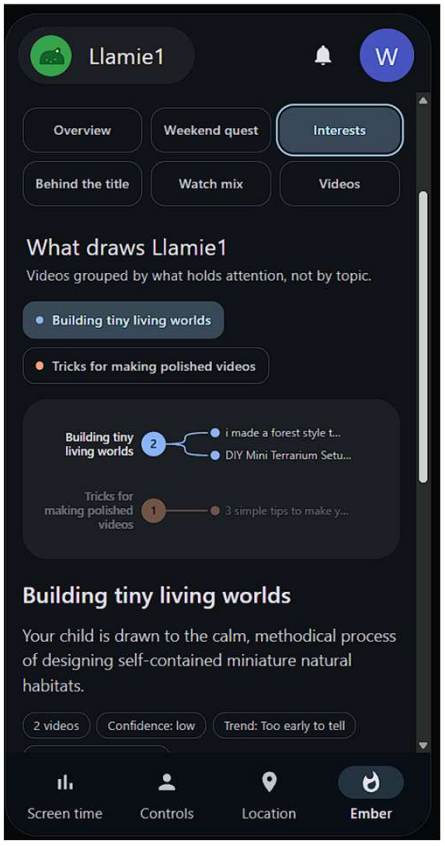


Snapshots of the prototype

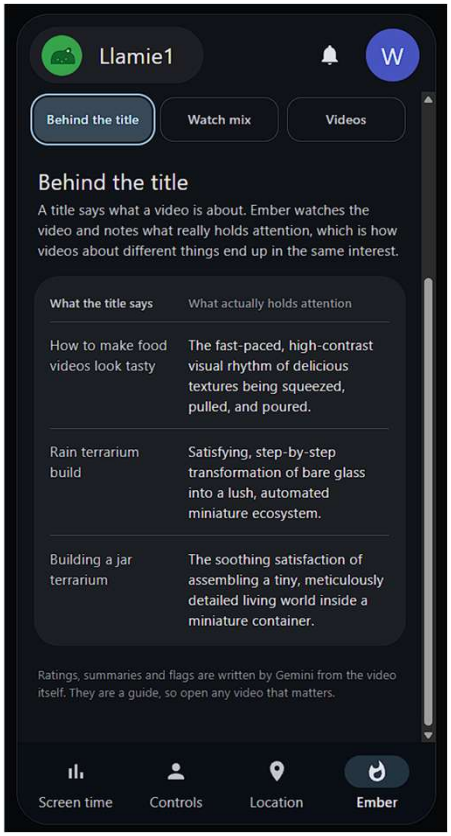


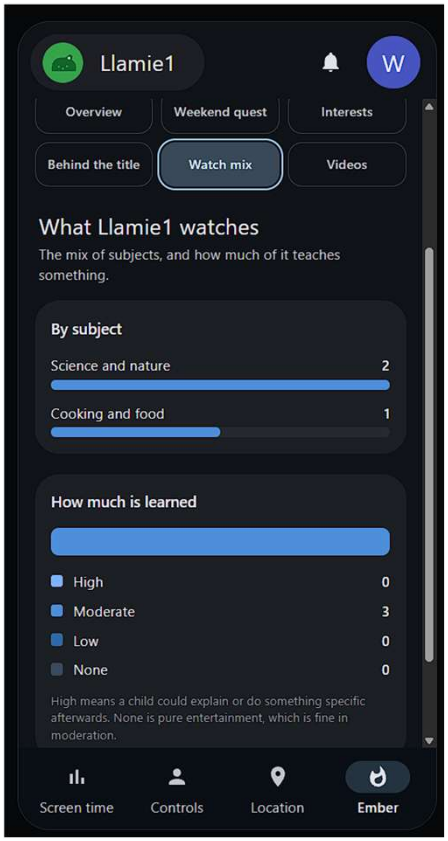


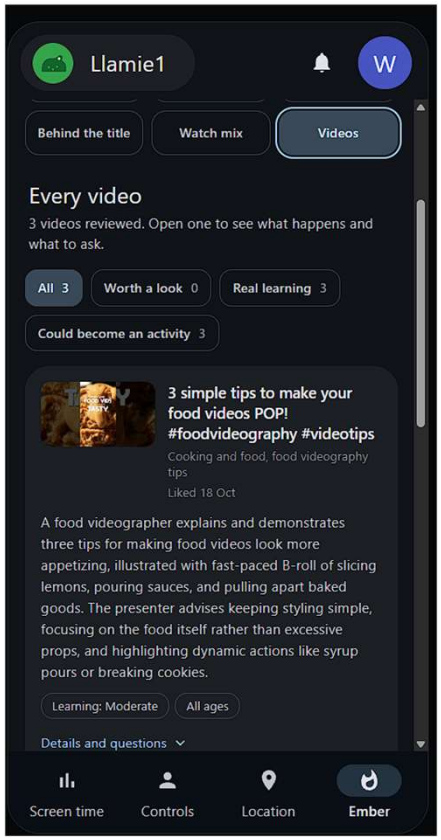




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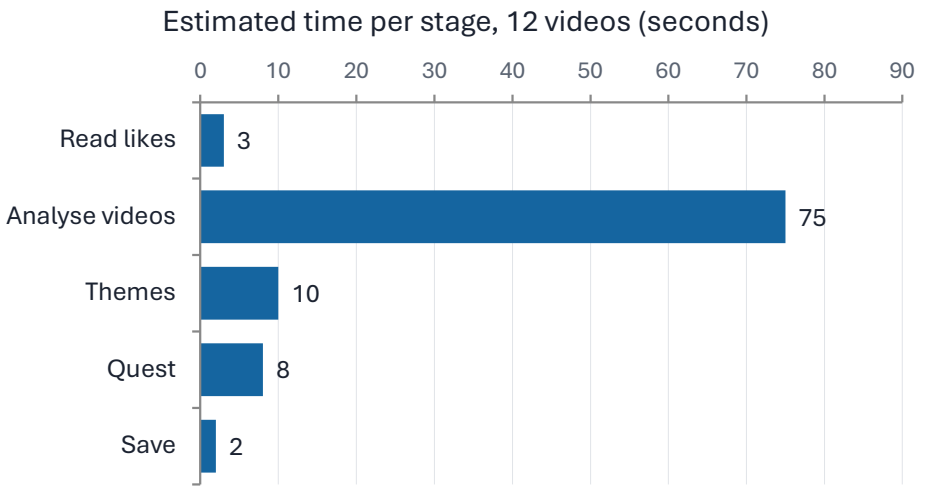
Prototype Performance report

Parameter	Assumption
Workload	12 liked videos per sync, about 8 minutes each
Gemini analysis	About 25 s per video, on a paid-tier quota
Concurrency	4 parallel tasks in the Cloud Run job
Fixed stages	Read likes 3 s, themes 10 s, quest 8 s, save 2 s
Report API	Three Datastore reads of about 20 ms each when warm
Cold start	1 to 2 s; avoided with one minimum instance
Repeat sync	Only new likes analysed; cached videos skipped

98 s
first sync, 12 videos

48 s
repeat sync, 2 new likes

< 300 ms
p95 report API, warm



Additional details and future development

Tested

- Liked videos and subscriptions read through Google sign-in on a standard account
- Gemini found real themes across 17 liked videos, not tags
- Runs end to end: analysis, themes, quest, dashboard

Found

- YouTube blocks supervised (Family Link) accounts from third-party apps
- The API returns likes, not watch history
- So Ember starts with teens on their own accounts, and saved links for younger children

Next

- Watch history through Takeout or the Data Portability API
- Common Ground: siblings' interests combined into one shared activity
- Google verification and a child-privacy review before launch

Provide links to your:

- GitHub Public Repository

https://github.com/WJG01/Ember

- Demo Video Link (3 Minutes)

https://drive.google.com/file/d/1f1B121AFnsmCoLkdG11SUUQn_zzd9Pgk/view?usp=sharing

- Final Product Link

https://ember-api-4a4r4yvz2a-as.a.run.app/

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Thank you

